Characterize optimal dynamic disclosure with continuous quality and quality-dependent completion rates

ChatGPT (gpt-5.6-sol)*

Partial progress generated on 2026-08-24

Abstract

We present partial progress on an open problem from *Motivating Effort with Information about Future Rewards*.

1 The open problem

This paper addresses an open problem posed by Chang Liu in *Motivating Effort with Information about Future Rewards* [1] (Economics and Computation (EC), 2022), Page 5 (Table 1) and page 30 (Conclusion, Section 6).. The website catalogue entry is problem 17 (`W3207326278_p0`) of the Open Problems in Operations Research collection.

> *Determine (in closed form) an optimal dynamic information policy σ for the principal in the above model when task quality q is continuously distributed (with prior F) and the completion rate depends on quality (λ_q varies with q). Equivalently, characterize the principal-optimal disclosure rule over time—including when and how much information about q to reveal—that maximizes the principal's expected discounted completion payoff subject to the agent's optimal stopping (quitting) incentives given belief updating from both disclosures and endogenous time information.*

The remainder of this document is the partial progress produced by the model, reproduced verbatim from the pipeline output.

Status: *partial progress relative to the requested primitive closed form.* The strong-duality, measurable-selection, lower-semicontinuity, and continuous-time obedience gaps can all be repaired. What remains unresolved is the elimination of the infinite-dimensional multiplier and, for $r_p \le r$, the determination of feasible mixing weights directly from primitives.

Assume throughout that quality lies in a standard Borel subset of $[0, \infty)$, that F is atomless, that q and $b(q)$ are integrable, that $\lambda(q) > 0$ and $b(q) > 0$ are measurable, and that $c, r, r_p > 0$.

*Generated by the `base_solver_ni_two_iterations` pipeline (Codex CLI, non-interactive; reasoning effort `ultra`; stages A.6 solve → A.8 verify → A.10 repair write-up; run `run_20260824_053314_3efc8d64`, `ec-2it-ultra`); two-iteration run, outcome from iteration 2. Independent verifier assessment: REJECTED with progress score 1/3 (minor partial progress). Maintained by the AI in Mathematics & Operations Research project.

Conditional on q, the policy's randomization is independent of $X \sim \mathrm{Exp}(\lambda(q))$. Weak obedience is used: when indifferent, the agent follows the recommendation.

Put

$$a(q) = \lambda(q)q - c, \qquad \rho(q) = r + \lambda(q),$$
$$p(q) = \lambda(q)b(q), \qquad \kappa(q) = r_p + \lambda(q).$$

For a deadline $D \in [0, \infty]$, define

$$u(q, D) = \frac{a(q)}{\rho(q)}\big(1 - e^{-\rho(q)D}\big), \qquad v(q, D) = \frac{p(q)}{\kappa(q)}\big(1 - e^{-\kappa(q)D}\big).$$

2 Exact obedience, including arbitrary stopping deviations

Let K_q be the conditional law of the pre-sampled deadline, and set

$$S(q, t) = K_q\big((t, \infty]\big).$$

At a continue history at time t, put

$$Z(t) = \int e^{-\lambda(q)t} S(q, t)\, F(dq).$$

Following all future recommendations yields the time-t value

$$\frac{e^{rt}}{Z(t)}\, C_K(t),$$

where

$$C_K(t) = \int a(q) \int_t^\infty e^{-\rho(q)u} S(q, u)\, du\, F(dq).$$

Hence the continue-obedience condition is

$$C_K(t) \geq 0 \qquad \forall t \geq 0. \tag{C}$$

If $Z(t) = 0$, then $S(q, t) = 0$ almost surely and therefore $S(q, u) = 0$ for every $u \geq t$; consequently $C_K(t) = 0$. Thus imposing (C) at unreachable continue histories adds no restriction.

Finite quit signals have joint measure

$$\nu(dq, dt) = e^{-\lambda(q)t}\, F(dq)\, K_q(dt).$$

Because both spaces are standard Borel and ν is finite, there is a disintegration

$$\nu(dq, dt) = \pi_t^Q(dq)\, m_Q(dt).$$

If a quit signal at t is ignored for an additional horizon h, the conditional payoff is

$$f_t(h) = \int \frac{a(q)}{\rho(q)}\big(1 - e^{-\rho(q)h}\big)\, \pi_t^Q(dq).$$

The quit-obedience condition is therefore

$$f_t(h) \leq 0 \quad \forall h \in [0, \infty], \quad \text{for } m_Q\text{-a.e. } t. \tag{Q}$$

The null set in (Q) can be chosen independently of h. For each fixed rational $h \geq 0$, obedience gives the inequality outside an m_Q-null set. Taking the union over rational h, and then using continuity of $h \mapsto f_t(h)$, including its limit as $h \to \infty$, yields one common full-measure set on which the inequality holds for every h.

Equivalently, (Q) can be written without selecting posteriors:

$$\int_Q \int_A e^{-\lambda(q)t} \frac{a(q)}{\rho(q)} \big(1 - e^{-\rho(q)h}\big)\, K_q(dt)\, F(dq) \leq 0 \tag{Q'}$$

for every Borel $A \subset [0, \infty)$ and every $h \in [0, \infty]$. Indeed, for fixed h, this signed measure has Radon–Nikodym derivative $f_t(h)$ with respect to m_Q.

These conditions are also sufficient against every continuous-time stopping deviation. To see this, condition first on the agent's private randomization. Along a path on which neither completion nor a quit signal has yet occurred, the observable history evolves deterministically. Therefore any pure stopping strategy has a deterministic first time s at which it would quit while recommendations remain "continue." If a quit signal arrives earlier, at $d < s$, then, on the no-completion path following that signal, the strategy selects some additional deterministic horizon $h(d)$. Progressive private randomization is merely a mixture of these pure plans.

By (Q), replacing every post-signal horizon $h(d)$ by immediate quitting weakly raises the payoff. The remaining deviation stops at s if no earlier quit signal arrives. Conditional on a continue history at $t \leq s$, the payoff difference between following the policy and this deviation is exactly

$$\frac{e^{rt}}{Z(t)}\, C_K(s) \geq 0.$$

Mixtures preserve the inequality. Thus (C)–(Q) are necessary and sufficient against arbitrary adapted stopping strategies, not only deterministic or infinitesimal deviations.

3 Exact reduction to an attained kernel program

Define

$$\mathcal{G} = \{q : a(q) \geq 0\}, \qquad \mathcal{B} = \{q : a(q) < 0\}.$$

Replacing K_q by δ_∞ on $\mathcal{G}$:

- weakly increases every $C_K(t)$;
- removes nonnegative contributions from (Q');
- weakly raises the principal's payoff, strictly wherever $p(q) > 0$ and the previous deadline was finite.

Hence an optimum may, and under $p > 0$ must, satisfy

$$K_q = \delta_\infty \quad F\text{-a.e. on } \mathcal{G}. \tag{1}$$

All finite quit signals then involve only strict-bad types, so (Q) is automatic.

For $q \in \mathcal{B}$, put

$$d(q) = c - \lambda(q)q > 0.$$

Define

$$R(t) = \int_{\mathcal{G}} \frac{a(q)}{\rho(q)}\, e^{-\rho(q)t}\, F(dq),$$

$$\ell_t(q, D) = d(q) \int_t^D e^{-\rho(q)u}\, du\, \mathbf{1}_{\{D>t\}}.$$

Then

$$C_K(t) = R(t) - L_K(t), \qquad L_K(t) = \int_{\mathcal{B}} \int \ell_t(q, D)\, K_q(dD)\, F(dq).$$

Thus the problem is exactly

$$\begin{aligned} P_{\mathcal{B}} = \max_K \quad & J(K) \coloneqq \int_{\mathcal{B}} \int v(q, D)\, K_q(dD)\, F(dq), \\ \text{s.t.} \quad & L_K(t) \le R(t) \qquad \forall t \ge 0. \end{aligned} \tag{P}$$

The full value is

$$V^* = V_{\mathcal{G}} + P_{\mathcal{B}}, \qquad V_{\mathcal{G}} = \int_{\mathcal{G}} \frac{p(q)}{\kappa(q)}\, F(dq).$$

Program (P) attains its maximum. Equip $[0, \infty]$ with its one-point-compactification topology. For every q and t, both $D \mapsto v(q, D)$ and $D \mapsto \ell_t(q, D)$ are continuous, including at $D = \infty$, and

$$0 \le v(q, D) \le b(q), \qquad 0 \le \ell_t(q, D) \le \frac{c}{r}.$$

The specific Young-measure compactness result needed here is the following: kernels from a finite standard Borel measure space into a compact metric action space form a compact set in the stable topology generated by the functionals

$$K \longmapsto \iint \psi(q, D)\, K_q(dD)\, F(dq), \qquad \psi \in L^1\big(F; C([0, \infty])\big).$$

One direct proof chooses countable dense families in $C([0, \infty])$ and $L^1(F)$, extracts weak-$*$ limits of the corresponding bounded kernel moments, and reconstructs the limiting probability kernels pointwise by the Riesz representation theorem. The objective and each constraint are continuous in this topology. The feasible set is therefore a nonempty closed subset of a compact set.

Moreover,

$$R - L_K \in C_0\big([0, \infty)\big).$$

Indeed, continuity follows by dominated convergence, while

$$0 \le L_K(t) \le \frac{c}{r}\, e^{-rt}, \qquad 0 \le R(t) \le e^{-rt} \int q\, F(dq).$$

4 Strong duality

Let $\mathcal{M}_+^f$ denote the finite positive Radon measures on $[0, \infty)$, equipped with the weak-$*$ topology induced by $C_0([0, \infty))$. For $\alpha \in \mathcal{M}_+^f$, define

$$\Phi_\alpha(q, D) = \int \ell_t(q, D)\, \alpha(dt), \qquad H_\alpha(q, D) = v(q, D) - \Phi_\alpha(q, D).$$

The Lagrangian is

$$\mathcal{L}(K,\alpha) = J(K) + \int \big(R(t) - L_K(t)\big)\,\alpha(dt).$$

For fixed α, the map $\mathcal{L}(\cdot,\alpha)$ is continuous and affine in the stable topology. This follows because

$$(q,D) \longmapsto \Phi_\alpha(q,D)$$

is measurable in q, continuous in D, and bounded by $(c/r)\,\alpha([0,\infty))$. For fixed K, the map $\mathcal{L}(K,\cdot)$ is continuous and affine in the weak-$*$ topology because $R - L_K \in C_0$.

The needed form of Sion's minimax theorem states that if one variable ranges over a compact convex subset of a locally convex space, the other over a convex set, and the function is upper-semicontinuous concave in the first variable and lower-semicontinuous convex in the second, then

$$\max_K \inf_\alpha \mathcal{L}(K,\alpha) = \inf_\alpha \max_K \mathcal{L}(K,\alpha).$$

All hypotheses have just been verified.

If K is feasible, then

$$\inf_{\alpha\in\mathcal{M}_+^f} \mathcal{L}(K,\alpha) = J(K),$$

with the infimum attained at $\alpha = 0$. If $L_K(t_0) > R(t_0)$, then choosing $\alpha = n\delta_{t_0}$ and letting $n \to \infty$ makes the infimum equal to $-\infty$. Hence

$$P_{\mathcal{B}} = \inf_{\alpha\in\mathcal{M}_+^f} \left\{ \int R\,d\alpha + \sup_K \int_{\mathcal{B}} \int H_\alpha(q,D)\,K_q(dD)\,F(dq) \right\}. \tag{2}$$

For finite α, H_α is a Carathéodory integrand: it is measurable in q, continuous in D, and integrably bounded. The measurable maximum theorem therefore yields a measurable maximizer $D_\alpha(q)$ and

$$\sup_K \int_{\mathcal{B}} \int H_\alpha(q,D)\,K_q(dD)\,F(dq) = \int_{\mathcal{B}} \max_D H_\alpha(q,D)\,F(dq). \tag{3}$$

This repairs the missing interchange step.

Now enlarge the multiplier space to

$$\mathcal{A}_R = \left\{ \alpha \ge 0 : \alpha \text{ locally finite Radon}, \quad \int R\,d\alpha < \infty \right\}.$$

Writing $M_\alpha(u) = \alpha([0,u])$, Tonelli's theorem gives, for finite D,

$$\Phi_\alpha(q,D) = d(q) \int_0^D e^{-\rho(q)u} M_\alpha(u)\,du,$$

and therefore

$$H_\alpha(q,D) = \int_0^D \Big[p(q)e^{-\kappa(q)u} - d(q)e^{-\rho(q)u} M_\alpha(u) \Big]\,du. \tag{4}$$

At $D = \infty$, define $H_\alpha(q,\infty)$ as the finite-D limit, possibly $-\infty$.

For every feasible K and every $\alpha \in \mathcal{A}_R$,

$$\int \Phi_\alpha\,d(KF) = \int L_K\,d\alpha \le \int R\,d\alpha < \infty.$$

Consequently,

$$J(K) \le \int R\, d\alpha + \int_{\mathcal{B}} \sup_{D} H_\alpha(q,D)\, F(dq).$$

Thus weak duality holds for all $\alpha \in \mathcal{A}_R$. Since finite measures are contained in $\mathcal{A}_R$, (2)–(3) imply the exact generalized dual

$$\boxed{P_{\mathcal{B}} = \inf_{\alpha \in \mathcal{A}_R} \left\{ \int R\, d\alpha + \int_{\mathcal{B}} \sup_{D \in [0,\infty]} H_\alpha(q,D)\, F(dq) \right\}.} \tag{D}$$

Dual attainment

Suppose $F(a>0)>0$. Then $R(t)>0$ for every finite t. Let (α_n) be a minimizing sequence in a bounded dual sublevel set. Since $H_{\alpha_n}(q,0)=0$,

$$\int R\, d\alpha_n \le C.$$

For every $T<\infty$,

$$\alpha_n\big([0,T]\big) \le \frac{C}{\min_{0\le t\le T} R(t)}.$$

A diagonal vague-compactness argument yields a locally finite Radon measure α and a subsequence with

$$\alpha_n \longrightarrow \alpha \quad \text{vaguely on every compact interval.}$$

Because $R \ge 0$ is continuous and vanishes at infinity,

$$\int R\, d\alpha \le \liminf_n \int R\, d\alpha_n.$$

For fixed q and finite D, the map $t \mapsto \ell_t(q,D)$ is continuous with compact support, so

$$H_{\alpha_n}(q,D) \longrightarrow H_\alpha(q,D).$$

The missing $D=\infty$ step is as follows. As $D\to\infty$, $v(q,D)$ converges and $\Phi_\alpha(q,D)$ increases to its limit by monotone convergence. Hence $H_\alpha(q,D)$ has a limit in $[-\infty,\infty)$, and

$$\sup_{D\in[0,\infty]} H_\alpha(q,D) = \sup_{D<\infty} H_\alpha(q,D). \tag{5}$$

The value at $D=\infty$, being a limit of finite-D values, cannot exceed their supremum. Therefore

$$\sup_D H_\alpha(q,D) \le \liminf_n \sup_D H_{\alpha_n}(q,D).$$

The supremum is measurable, because finite-D continuity allows the supremum in (5) to be taken over nonnegative rational D. Since it lies between 0 and $b(q)$, Fatou's lemma applies. Thus the generalized dual objective is lower-semicontinuous, and α is a dual optimizer.

This completes the duality and dual-attainment repair.

If $F(a>0)=0$, then $R\equiv 0$. The $t=0$ constraint becomes

$$\int_{\mathcal{B}} \int \frac{d(q)}{\rho(q)} \big(1-e^{-\rho(q)D}\big)\, K_q(dD)\, F(dq) \le 0,$$

so

$$D = 0 \quad \text{a.s. on } \mathcal{B}.$$

Neutral types ($a=0$) may be assigned $D=\infty$, which is principal-optimal and obedient under weak tie-breaking.

5 Exact optimality certificates

Let K^* and α^* be primal and dual optimizers. Put

$$s_{\alpha^*}(q) = \sup_D H_{\alpha^*}(q, D).$$

The primal–dual gap has the exact decomposition

$$\begin{aligned} 0 = &\int \big(R(t) - L_{K^*}(t)\big)\, \alpha^*(dt) \\ &+ \int_{\mathcal{B}} \int \big[s_{\alpha^*}(q) - H_{\alpha^*}(q, D)\big]\, K_q^*(dD)\, F(dq). \end{aligned} \tag{6}$$

Both terms are nonnegative. Hence

$$R(t) = L_{K^*}(t) \quad \alpha^*\text{-a.e.}, \tag{7}$$

and

$$K_q^*\big(\operatorname{Argmax}_D H_{\alpha^*}(q, D)\big) = 1 \quad F\text{-a.e. } q \in \mathcal{B}. \tag{8}$$

Conversely, any feasible K and any $\alpha \in \mathcal{A}_R$ satisfying (7)–(8) attain equality in weak duality and are therefore optimal. These conditions are thus necessary and sufficient for global optimality.

6 What the characterization gives in the three discount regimes

For $q \in \mathcal{B}$, define

$$I(q) = \frac{p(q)}{d(q)} = \frac{\lambda(q) b(q)}{c - \lambda(q) q}, \qquad \delta = r_p - r.$$

Then

$$H_\alpha(q, D) = d(q) \int_0^D e^{-\rho(q) u} \Big[I(q) e^{-\delta u} - M_\alpha(u)\Big]\, du. \tag{9}$$

The case $r_p > r$

Set

$$h_\alpha(t) = e^{\delta t} M_\alpha(t).$$

This function is nondecreasing; moreover, once positive it is strictly increasing, because $e^{\delta t}$ is strictly increasing and M_α is nondecreasing. For $I(q) > 0$, define

$$D^*(q) = \inf\{t : h_{\alpha^*}(t) \geq I(q)\}.$$

The derivative of $H_{\alpha^*}(q, D)$ has the sign of $I(q) - h_{\alpha^*}(D)$; hence the global maximizer is unique and

$$\boxed{D^*(q) = \inf\left\{t \geq 0 : e^{(r_p - r)t} M_{\alpha^*}(t) \geq \frac{\lambda(q) b(q)}{c - \lambda(q) q}\right\}.} \tag{10}$$

At a multiplier atom,

$$M_{\alpha^*}(D^-) \leq I(q) e^{-(r_p - r) D} \leq M_{\alpha^*}(D). \tag{11}$$

Writing $M = M_{\alpha^*}$, feasibility and complementary slackness become

$$W(t) \coloneqq R(t) - \int_t^\infty \int_{\{q\in\mathcal{B}:\, I(q)>e^{\delta u}M(u)\}} d(q)e^{-\rho(q)u}\, F(dq)\, du \geq 0,$$
$$W(t) = 0 \qquad dM\text{-a.e.} \tag{12}$$

This is an exact nonlinear complementarity system, but it still has to be solved for M.

The case $r_p = r$

Now M itself is nondecreasing. Define

$$D_-(q) = \inf\{t : M(t) \geq I(q)\}, \qquad D_+(q) = \inf\{t : M(t) > I(q)\}.$$

Then

$$\operatorname{Argmax}_D H_{\alpha^*}(q, D) = [D_-(q), D_+(q)]. \tag{13}$$

A primal optimizer may mix on this interval. The support restriction (13), feasibility, and complementary slackness determine the admissible aggregate mixing, but they do not provide its weights in closed form.

The case $r_p < r$

Put $\gamma = r - r_p > 0$. Then

$$H_{\alpha^*}(q, D) = d(q) \int_0^D e^{-\rho(q)u} \left[I(q)e^{\gamma u} - M_{\alpha^*}(u)\right] du. \tag{14}$$

Both $I(q)e^{\gamma u}$ and $M_{\alpha^*}(u)$ are nondecreasing, so their difference can change sign repeatedly. A first-crossing rule is therefore invalid. The exact conclusion is only

$$K_q^*\left(\operatorname{Argmax}_{D\in[0,\infty]} \int_0^D e^{-\rho(q)u} \left[I(q)e^{\gamma u} - M_{\alpha^*}(u)\right] du\right) = 1. \tag{15}$$

Feasibility and (7) must then determine which global maximizers are used, and with what weights.

The implemented disclosure policy samples $D \sim K_q^*$, independently of completion conditional on q, recommends continuation while $t < D$, and recommends quitting at D. Its continuation posterior is

$$\pi_t^C(dq) = \frac{e^{-\lambda(q)t} S^*(q, t)\, F(dq)}{\int e^{-\lambda(z)t} S^*(z, t)\, F(dz)}.$$

Thus completion-rate learning is retained exactly.

As a boundary check, universal continuation $D = \infty$ is feasible, and therefore optimal, exactly when

$$\int \frac{a(q)}{\rho(q)}\, e^{-\rho(q)t}\, F(dq) \geq 0 \qquad \forall t \geq 0. \tag{16}$$

7 Sharpness: irregular rules persist with genuinely varying bad-type rates

The earlier arbitrary-deadline example can be strengthened so that completion rates already vary among strict-bad types.

Suppose $r_p > r$. Take two atomless strict-bad blocks B_1, B_2 with distinct constant rates $\lambda_1 \neq \lambda_2$, and choose their qualities so that $\lambda_j q < c$. Let $\tau : B_1 \cup B_2 \to [0,T]$ be any bounded measurable function. For $q \in B_j$, write

$$d(q) = c - \lambda_j q, \quad \rho_j = r + \lambda_j, \quad x(q) = 1 - e^{-\rho_j \tau(q)},$$

and set

$$b(q) = \frac{d(q)}{\lambda_j} \, e^{(r_p - r)\tau(q)}.$$

Add an atomless good block G_j with rate λ_j, and choose its mass and qualities so that

$$\int_{G_j} (\lambda_j q - c) \, F(dq) = \int_{B_j} d(q)x(q) \, F(dq). \tag{17}$$

This can be done with arbitrarily small good-type mass by placing G_j at sufficiently high qualities. Fill any remaining mass with atomless neutral types.

For the proposed deadline $D = \tau(q)$,

$$R(t) = \sum_j \frac{e^{-\rho_j t}}{\rho_j} \int_{B_j} d(q)x(q) \, F(dq),$$

and

$$L(t) \leq R(t)$$

because

$$\left(e^{-\rho_j t} - e^{-\rho_j \tau}\right)_+ \leq e^{-\rho_j t}\left(1 - e^{-\rho_j \tau}\right).$$

The time-zero constraint binds.

For any competing deadline D, let $z = 1 - e^{-\rho_j D}$. On B_j,

$$v(q, D) = \frac{p(q)}{\kappa_j} \left[1 - (1 - z)^{\kappa_j/\rho_j}\right], \qquad \kappa_j = r_p + \lambda_j.$$

Since $\kappa_j/\rho_j > 1$, this expression is strictly concave in z. Its derivative at $z = x(q)$ equals

$$\frac{p(q)}{\rho_j}\left(1 - x(q)\right)^{(r_p - r)/\rho_j} = \frac{d(q)}{\rho_j}.$$

Therefore the tangent inequality and the time-zero resource constraint give

$$\int v(q, D) \, F(dq) \leq \int v\big(q, \tau(q)\big) \, F(dq),$$

with equality only when $z = x(q)$, hence $D = \tau(q)$, almost surely. The dual certificate is $\alpha = \delta_0$.

Thus, even when completion rates differ among strict-bad types, the unique optimal deadline can be an arbitrary bounded measurable function of quality. Universal quality cutoffs, monotonicity, and finite cascades are therefore false.

There is a corresponding construction for $r_p \le r$. Let the weighted deadline distribution on each bad block satisfy

$$\frac{d(q)\,F(dq)}{A_j} \circ \tau^{-1} = \operatorname{Exp}(\eta_j), \qquad A_j = \int_{B_j} d(q)\,F(dq).$$

Then

$$L_j(t) = \frac{A_j}{\rho_j + \eta_j}\, e^{-(\rho_j+\eta_j)t}.$$

Add a good block with completion rate $\lambda_j + \eta_j$ and total agent surplus A_j, so that $R_j = L_j$. Put

$$\gamma = r - r_p \ge 0, \qquad g(t) = 2 - e^{-t}, \qquad M(t) = e^{\gamma t} g(t),$$

and set

$$I(q) = g\bigl(\tau(q)\bigr), \qquad b(q) = \frac{d(q)I(q)}{\lambda_j}.$$

The Stieltjes measure with cumulative function M belongs to $\mathcal{A}_R$, because

$$\rho_j + \eta_j - \gamma = r_p + \lambda_j + \eta_j > 0.$$

Moreover,

$$\frac{\partial H_\alpha(q,D)}{\partial D} = d(q)e^{-(\rho_j-\gamma)D}\bigl[g\bigl(\tau(q)\bigr) - g(D)\bigr],$$

so $D = \tau(q)$ is the unique global maximizer. Since $R = L$ for every t, complementary slackness proves optimality. Measure-preserving rearrangements of the exponential deadlines can make $\tau(q)$ arbitrarily nonmonotone and irregular, while $\lambda_1 \ne \lambda_2$ gives genuine nonstationarity among strict-bad types.

Exact disposition of the rejection reasons

- **Strong duality:** fully repaired by the finite-measure Sion argument, followed by the weak-duality extension to locally finite multipliers.
- **Measurable maximum and $D = \infty$ lower semicontinuity:** fully repaired by the Carathéodory measurable-maximum theorem and identity (5).
- **Continuous-time necessity and sufficiency of (C)–(Q):** fully repaired by the common-null-set construction and the explicit first-deviation reduction for arbitrary stopping strategies.
- **Primitive closed form for the full problem:** not repaired. Even for $r_p > r$, the function M must solve the infinite-dimensional complementarity system (12). For $r_p = r$, plateau mixing must additionally be chosen; for $r_p < r$, global maximizers and their feasible weights must be determined jointly.

Accordingly, the strongest rigorously proved result is an attained primal program, an attained exact generalized dual, and necessary and sufficient primal–dual optimality conditions for every discount regime. A general policy expressed directly in primitives, with no remaining infinite-dimensional multiplier or mixing problem, remains a genuine gap.

References

[1] Chang Liu. *Motivating Effort with Information about Future Rewards*. Economics and Computation (EC), 2022. https://doi.org/10.1145/3490486.3538319